THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS · MEDIUM LEVEL CHAPTER TEST SET 3

Chapter 5: **Circles** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. $\angle AOB = 136^\circ$ at the centre and C lies on the major arc AB. Then $\angle ACB =$ 1

(a) 78° **(b)** 58° **(c)** 88° **(d)** 68°

2. $\angle AOB = 130^\circ$ at the centre and D lies on the minor arc AB. Then $\angle ADB =$ 1

(a) 135° **(b)** 115° **(c)** 105° **(d)** 125°

3. In a circle, a chord of length 14 cm is at distance 24 cm from the centre. Another chord of the same length is at a distance (in cm): 1

(a) 24 **(b)** 25 **(c)** 23 **(d)** 26

4. AB is a diameter and C lies on the circle with $\angle ABC = 47^\circ$. Then $\angle BAC =$ 1

(a) 63° **(b)** 53° **(c)** 43° **(d)** 33°

5. The length of a chord of a circle of radius 17 cm lying 15 cm from the centre is (in cm): 1

(a) 15 **(b)** 16 **(c)** 18 **(d)** 17

6. A chord of length 30 cm lies 8 cm from the centre of a circle. The radius (in cm) is: 1

(a) 18 **(b)** 19 **(c)** 16 **(d)** 17

7. In a circle with centre O, chord AB has $\angle AOB = 124^\circ$. Then $\angle OBA =$ 1

(a) 28° **(b)** 48° **(c)** 18° **(d)** 38°

8. The angles of a cyclic quadrilateral $ABCD$ at A and C are in the ratio $7:5$. Then $\angle A =$ 1

(a) 125° **(b)** 115° **(c)** 95° **(d)** 105°

9. $\angle AOB = 152^\circ$ at the centre and C lies on the major arc AB. Then $\angle ACB =$ 1

(a) 76° **(b)** 86° **(c)** 66° **(d)** 96°

10. $\angle AOB = 150^\circ$ at the centre and D lies on the minor arc AB. Then $\angle ADB =$ 1

(a) 125° **(b)** 95° **(c)** 115° **(d)** 105°

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. In a circle, $\angle AOB = 116^\circ$ with O the centre. Find $\angle OAB$ and $\angle ACB$ (C on the major arc). 2

12. Find the length of the chord of a circle of radius 29 cm that is 21 cm from the centre. 2

13. A cyclic quadrilateral $ABCD$ has $\angle A = 58^\circ$ and $\angle B = 91^\circ$. Find the angles $\angle C$ and $\angle D$. 2

14. In a circle, $\angle AOB = 144^\circ$ with O the centre. Find $\angle OAB$ and $\angle ACB$ (C on the major arc). 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Two parallel chords of lengths 10 cm and 24 cm lie on opposite sides of the centre of a circle of radius 13 cm. Find the distance between them. 3

16. In a cyclic quadrilateral $ABCD$, $\angle A = (3x + 18)^\circ$ and $\angle C = (3x - 6)^\circ$. Find x, $\angle A$ and $\angle C$. 3

17. O is the centre of a circle and $\angle AOB = 148^\circ$. A point C lies on the major arc and D on the minor arc. Find $\angle OAB$, $\angle ACB$ and $\angle ADB$. 3

OR

Two parallel chords of lengths 16 cm and 30 cm lie on opposite sides of the centre of a circle of radius 17 cm. Find the distance between them. 3

18. In a cyclic quadrilateral $ABCD$, $\angle A = (4x + 12)^\circ$ and $\angle C = (5x - 3)^\circ$. Find x, $\angle A$ and $\angle C$. 3

OR

O is the centre of a circle and $\angle AOB = 156^\circ$. A point C lies on the major arc and D on the minor arc. Find $\angle OAB$, $\angle ACB$ and $\angle ADB$. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. A chord AB of a circle with centre O is 48 cm long and the radius is 25 cm. Find the distance of the chord from O and the area of triangle OAB. 5

OR

Two chords of the same circle have lengths 12 cm and 16 cm and the circle has radius 10 cm. Find the distances of the chords from the centre and say which chord is nearer to the centre. 5

20. A chord AB of a circle with centre O is 48 cm long and the radius is 26 cm. Find the distance of the chord from O and the area of triangle OAB. 5

OR

Two chords of the same circle have lengths 6 cm and 8 cm and the circle has radius 5 cm. Find the distances of the chords from the centre and say which chord is nearer to the centre. 5

END OF QUESTION PAPER — Total: 40 Marks

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